

Inverse functions



1 Graph $y = x$ on the same axes as the graphs of the two functions and look to see one of the graphs is a reflection of the other about $y = x$. If they are reflections, then they are inverses.

2

a No

b Yes

c No

d Yes

e No

f Yes

g No

3

a $y = \frac{x+8}{7}$

b $y = \frac{3}{4}(x+5)$

c $y = \sqrt[3]{\frac{x}{6}}$

d $y = x^3 + 2$

e $y = \frac{8}{x-5} + 9$

f $y = \sqrt[3]{x-3} + 4$

g $y = \frac{8}{5x} - \frac{3}{5}$

h $y = \sqrt[3]{\frac{7}{x+3}} + 8$

i $y = \frac{\sqrt[3]{\frac{x+7}{4}} + 2}{9}$

4

a i $f^{-1}(x) = \frac{1}{8}(x+9)$

ii $[-41, 7]$

iii $[-4, 2]$

b i $f^{-1}(x) = \sqrt{x}$

ii $[0, \infty)$

iii $[0, \infty)$

c i $f^{-1}(x) = \sqrt{16-x^2}$

ii $[0, 4]$

iii $[0, 4]$

5

a $y = -\frac{4}{x}$

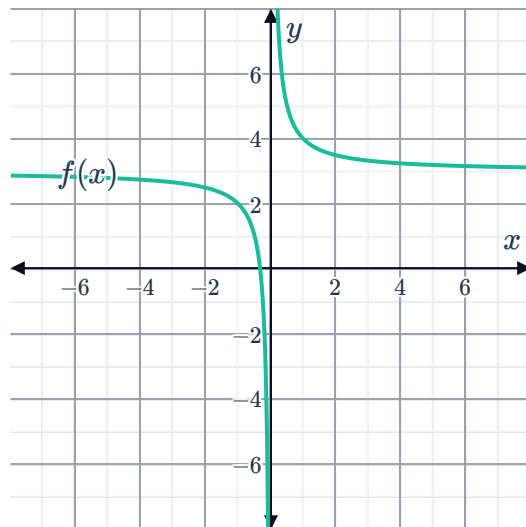
b $y = (x-5)^3$



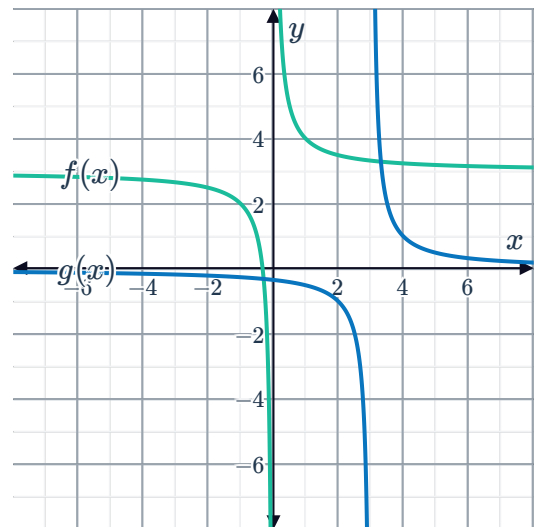
Graphs of inverse functions

6

a

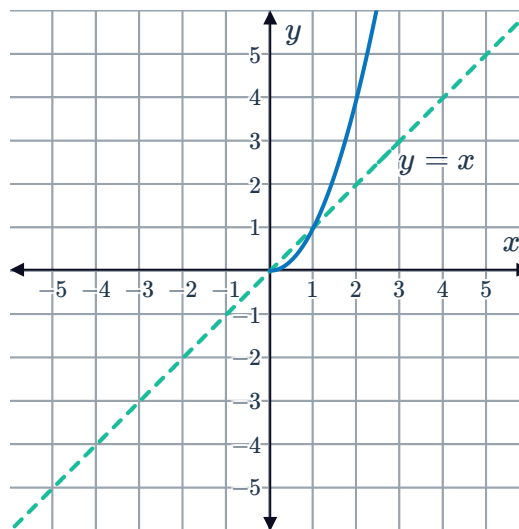


b



c Yes

7



8

a $x^3 - 2$

b $\sqrt[3]{x+2}$

c x

d x

e

i Yes

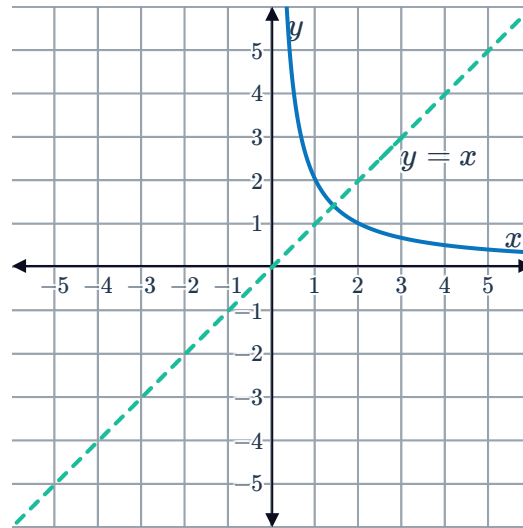
ii No

iii Yes

iv Yes

9

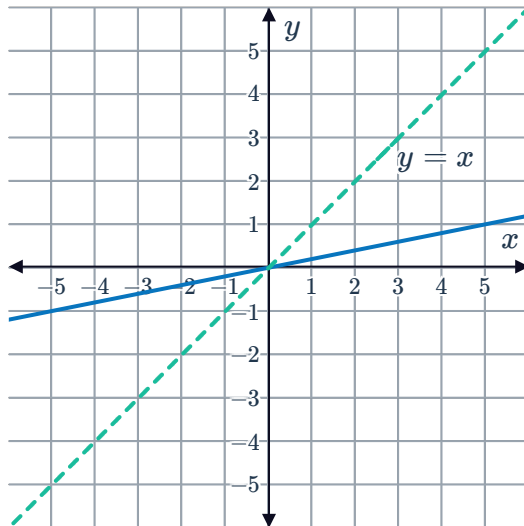
a



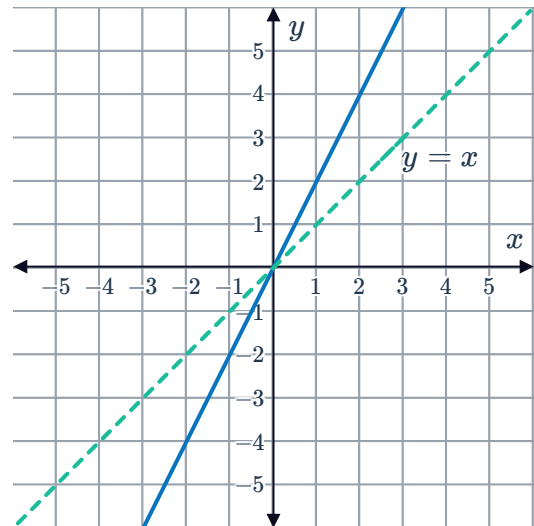
b The inverse graph is exactly the same as the original graph.

10

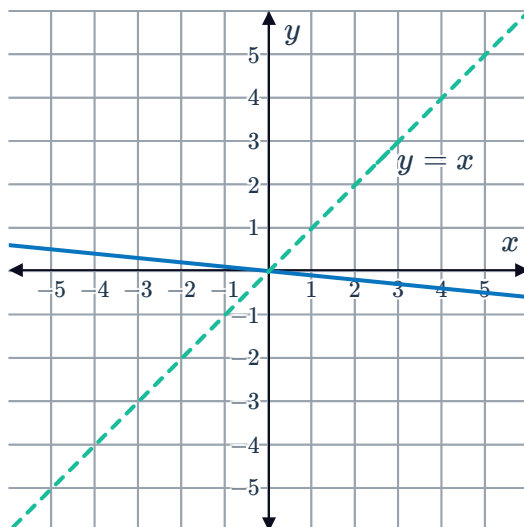
a



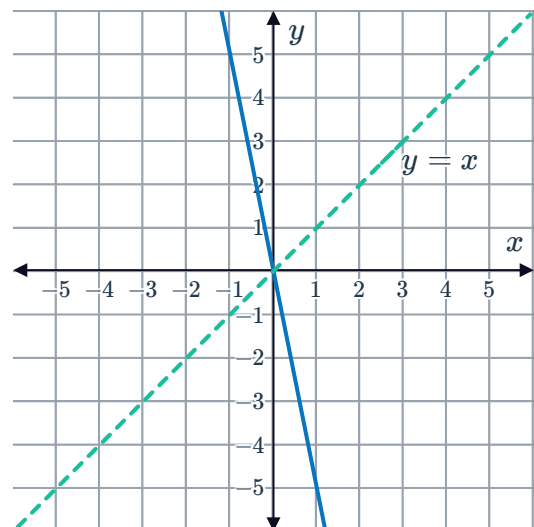
b



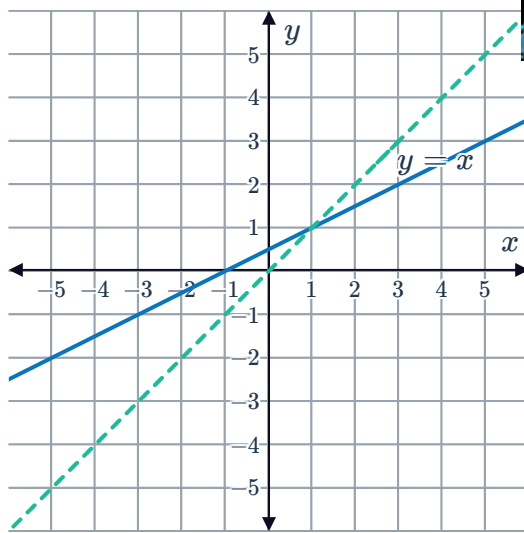
c



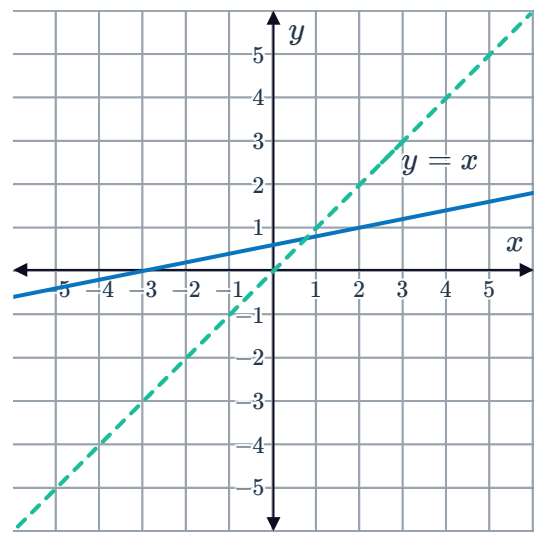
d



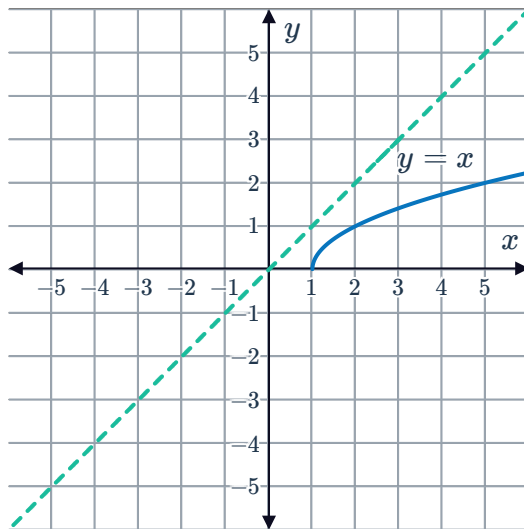
e



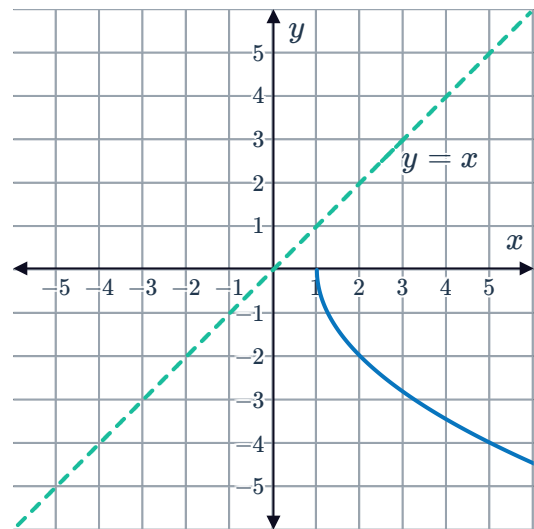
f



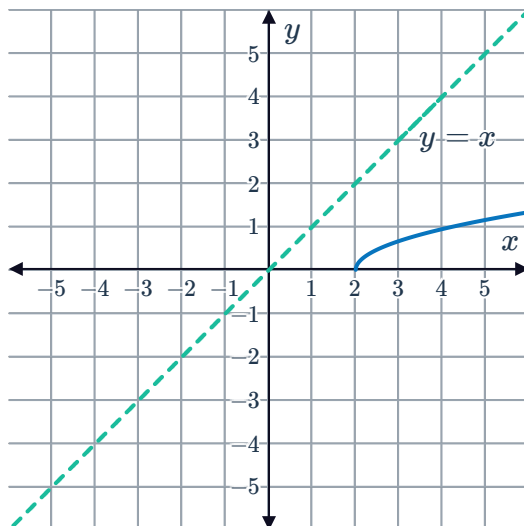
g



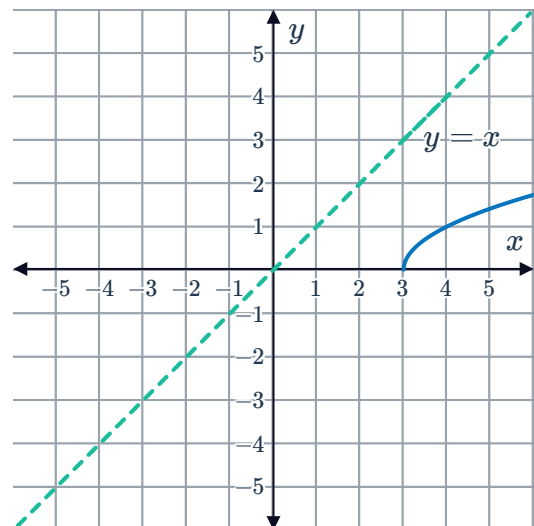
h



i



j



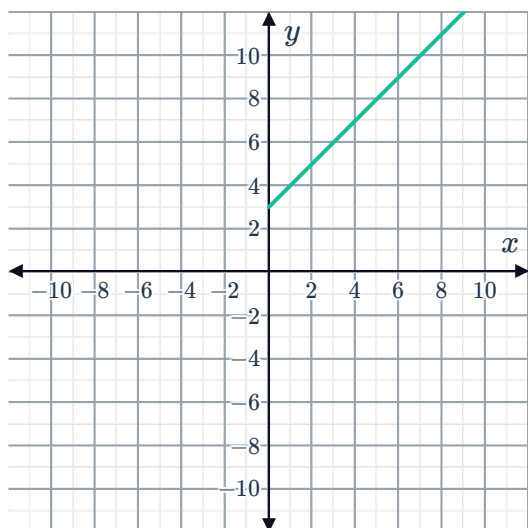
11

a

i



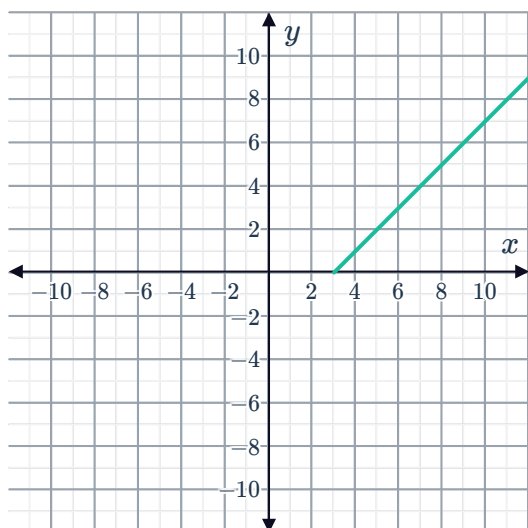
ii $f^{-1}(x) = x - 3$



iii $[3, \infty)$

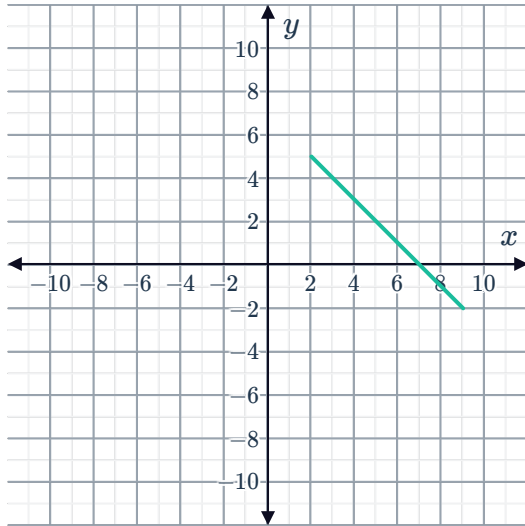
iv $[0, \infty)$

v



b

i



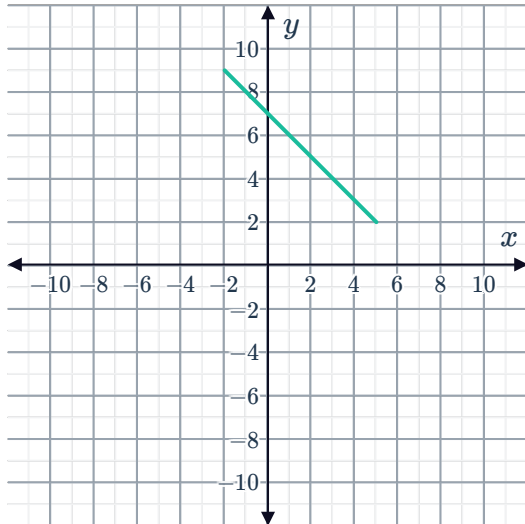
ii

$$f^{-1}(x) = 7 - x$$

iii $[-2, 5]$

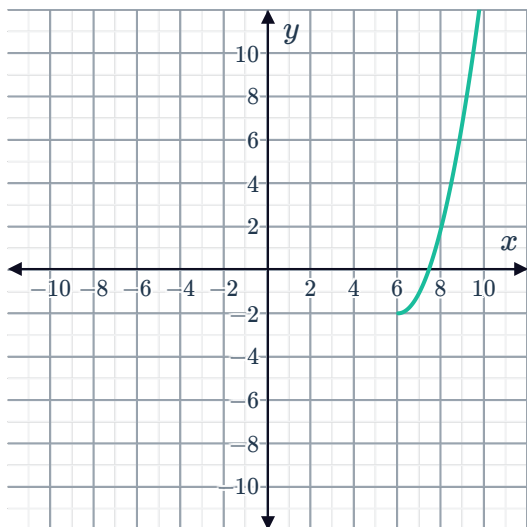
iv $[2, 9]$

v



c

i



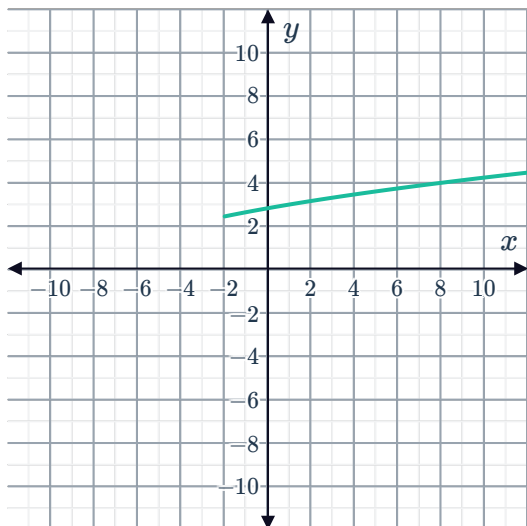
ii

$$f^{-1}(x) = \sqrt{x+2} + 6$$

iii $[-2, \infty)$

iv $[6, \infty)$

v



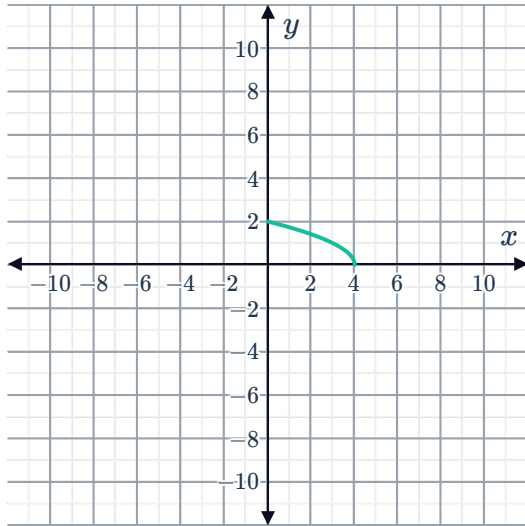
d

i



ii

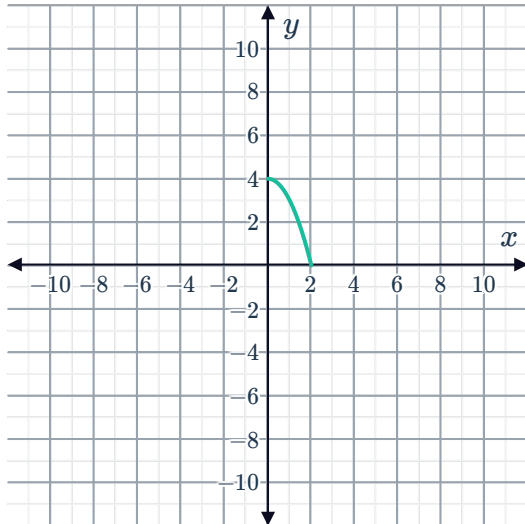
$$f^{-1}(x) = 4 - x^2$$



iii $(0, 2]$

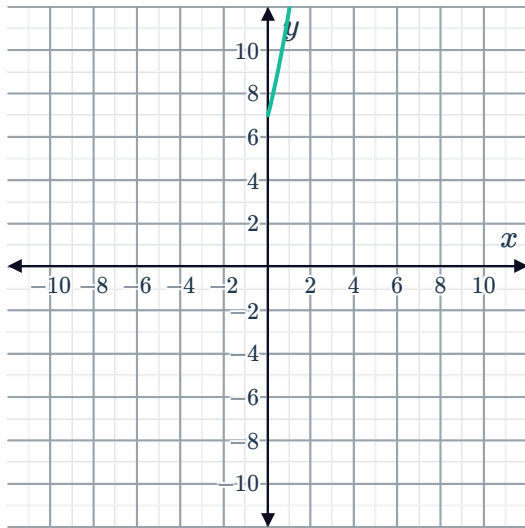
iv $[0, 4)$

v



e

i



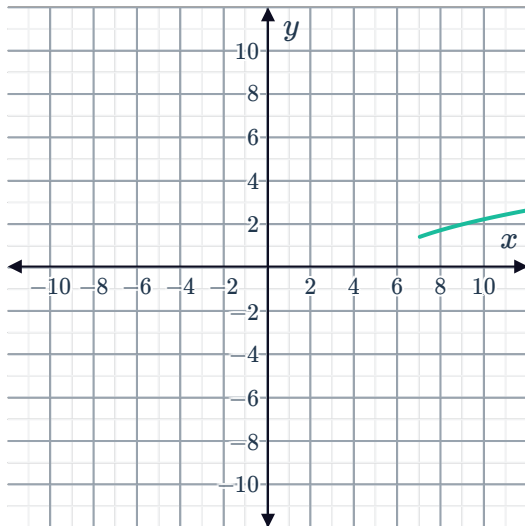
ii

$$f^{-1}(x) = \sqrt{x-3} - 2$$

iii $[7, \infty)$

iv $[0, \infty)$

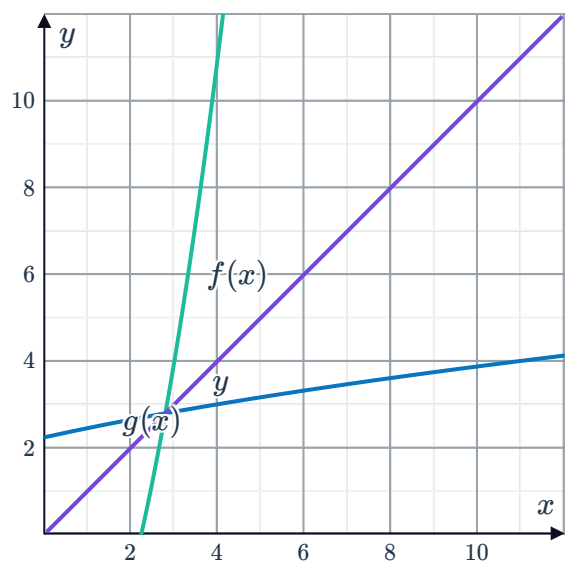
v



12

a $y = x$

b



c $f(x)$ is a reflection of $g(x)$ about $y = x$.



Applications

13

a $d = 4.9t^2$

b $d = 122.5$

14

a -25°

b 30°

c $T^{-1}(x) = \frac{9}{5}x + 32$

d For converting Celsius temperatures x to Fahrenheit temperatures $T^{-1}(x)$

15

a $t = \frac{5}{6}$

b $t = \frac{80 - \sqrt{v}}{96}$

16

a $T = 1.07A$

b $A = \frac{T}{1.07}$

17

a $l = \frac{9.8T^2}{4\pi^2}$

b $l = 0.6 \text{ m}$